

End semester exam -2023

MCA(III Sem.)

Software Engineering

Max. Marks: 100

Time 3.00 hours

1. What are the basic activities in the requirements process? By taking a suitable real life project, list the activities. (5+5)
 Introduction → Functional → Documenting customer need not understood properly
2. What are the most common errors in an SRS. Write at least two examples. (3+2)
3. A conference management site which allows authors to submit papers, program chairs to assign reviewers and do the final paper selection based on reviews, and reviewers to enter the review. List the major use cases for this system, along with the goals, preconditions, and exception scenarios. (10)
 Actor, Success Scenarios, Alternative way.
4. What is Effort Adjustment Factor (EAF)? List its uses. (5)
5. Why Agile methodology is popular now a days. List its merits and demerits. Briefly describe any two framework. (10)
 XP, SCRUM
6. What is the role of effort estimation in a project, and why is it important to do this estimation early? List some estimation models. (8)
 For effort, lost work, rework, feedback, rework box (loc), Black box (finds point earlier)
7. Suppose an organization plans to use COCOMO for effort estimation, but it wants to use only three cost drivers—product complexity, programmer capability, and development schedule. In this situation, from the initial estimate, by how much can the final estimate vary? (7)
 Effort = $a \times (LOC)^b \times PM \times Time = b_1 (Effort)^{b_2} \times M$
8. Suggest how you will evaluate a proposed architecture from a modifiability perspective. (5)
 Modularity
9. What is the cohesion of the following module? How would you change the module to increase cohesion? (8)
 Sequential, procedural, Functional cohesion

```

procedure file (file ptr, file name, op name);
begin
    case op name of
        "open": perform activities for opening the file.
        "close": perform activities for opening the file.
        "print": print the file
    end case
end
    
```

Communicational, Sequential, Udy!
10. Consider a program containing many modules. If a global variable x must be used to share data between two modules A and B, how would you design the interfaces of these modules to minimize coupling? (7)
 Data, address
11. Consider the following two algorithms for searching an element E in a sorted array A, which contains n integers. The first procedure implements a simple linear search algorithm. The second performs a binary search. Binary search is generally much more efficient in terms of execution time compared to the linear search.

```
function lin_search (A, E): boolean
```

```
var
```

```
  i : integer;
```

```
  found: boolean;
```

```
begin
```

```
  found := false;
```

```
  i := 1;
```

```
  while (not found) and (i ≤ n) do begin
```

```
    if (A[i] = E) then found := true;
```

```
    i := i + 1;
```

```
  end;
```

```
  lin_search := found;
```

```
end;
```

```
function bin_search (A, E): boolean
```

```
var
```

```
  low, high, mid, i, j : integer;
```

```
  found : boolean;
```

```
begin
```

```
  low := 1;
```

```
  high := n;
```

```
  found := false;
```

```
  while (low ≤ high) and (not found) do begin
```

```
    mid := (low + high)/2;
```

```
    if E < A[mid] then high := mid - 1
```

```
      else if E > A[mid] then low := mid + 1
```

```
      else found := true;
```

```
  end;
```

```
  bin_search := found;
```

```
end;
```

$$E = N + P$$

Determine the cyclomatic complexity for the two functions. (15)

12. What are the different levels of testing and the goals of these different levels? Briefly describe white box testing techniques and Black box testing technique. (10)

unit testy - unit

Integration -

system testy -

acceptance testy - user and

